

CASE 4

THE RENT UR CAR DATABASE

Designing a Relational Database to Create Tables, Forms, Queries, and Reports

PREVIEW

In this case, you will design a relational database for a new business that links private vehicle owners with customers who want to rent those vehicles. After your design is complete and correct, you will create database tables and populate them with data. Then you will produce one form with a subform, eight queries, and two reports. Two queries will display lists of high-end cars that are available for rental. Other queries will list the number of days each vehicle has been rented, the total amount of money each customer has spent on rentals, and the number of times each vehicle has been rented. Two more queries will display lists of the top-grossing vehicles and cars that have never been rented. You will also produce two reports based on queries that summarize customer rentals and rental income for vehicle owners.

PREPARATION

- Before attempting this case, you should have some experience in database design and in using Microsoft Access.
- Complete any part of Tutorial A that your instructor assigns.
- Complete any part of Tutorial B that your instructor assigns, or refer to the tutorial as necessary.
- Refer to Tutorial F as necessary.

BACKGROUND

The sharing economy is all the rage now, and many people are turning to alternatives for travel. For example, Airbnb now hosts many travelers who traditionally booked hotel rooms in the past. Car rental is undergoing a transformation as well; private vehicle owners now rent their cars to others when the vehicles are not in use. For example, if you commute to work using public transportation, your car is idle all day. Why not turn your vehicle into a source of extra income? Your cousin, Penny, did just that. She rarely uses her car because she lives in Chicago and takes public transportation everywhere, so she started to let her friends use her car for a daily fee. They made sure to return the car with the same amount of gasoline it contained when they borrowed it. Penny's friends liked the arrangement so much that they encouraged her to start a business for like-minded people who want to rent their cars. The business, called Rent Ur Car, also known as RUC, has taken off. Penny has been keeping records in Microsoft Excel spreadsheets and on paper, but she realizes that the business has outgrown her substandard record-keeping. Because you have been studying databases, she enlists your help over the summer to create a prototype database that will eventually be ported online and have its front end developed into an app.

On the first day of your summer job, you meet with Penny to understand how the business works. She describes its simple model: Owners register with RUC, submit their cars' details, and set a daily rental price. Customers also register with RUC and can book a rental car after browsing the types of vehicles available. Penny takes a small percentage of each transaction to pay herself, her team of workers (including you), and the insurance that covers each car rental. For this project, Penny wants you to focus on the rental aspect of her business and ignore the money she makes from each transaction.

Next, you examine her paper and spreadsheet records to better understand how to design your database. Penny has a listing of all the owners who have registered their cars for rental. The list is a stack of index cards; each contains an owner's name, address, phone number, and email address. The stack is alphabetized and organized in a box. In a separate box, Penny keeps a stack of cards that summarize the vehicles available for rent. Each vehicle's description is written on a card, including the price per day desired and a code that matches up with the owner. In an Excel spreadsheet, Penny has listed each vehicle, customer, and the start and end dates rented.

Although the database design appears fairly straightforward, Penny assures you that there are plenty of tasks to accomplish and questions to answer.

Eventually, you envision the app working this way: Customers can book a vehicle by choosing its ID number after perusing the table of vehicles and indicating their rental start and end dates. Because this app is a prototype built with Microsoft Access, you are confident that a form will test the future app's capabilities.

Penny tells you that many customers call and ask if specific vehicles are available. For example, she wants you to create a query that searches for and then displays a list of any Ferraris in the database. She also would like a query that searches for Corvettes available for rental and then lists the cities where the cars are located.

Penny is interested in analyzing data from the business and has a series of questions related to top performers that she wants you to organize in queries. First, she would like to know which vehicles are rented the most number of days. She is also curious to know who the top customers are, as measured by the amount of money each has spent on rentals. In addition, she wants to know how many times each vehicle has been rented, and she thinks that knowing which vehicles are the top-grossing rentals would be helpful for marketing.

Although Penny never turns down any owners who want to rent their vehicles, she knows that some vehicles are never rented. She wonders if you can create a query that can list these vehicles. You are confident that Microsoft Access is capable of this task.

Occasionally, Penny feels that prices need to be adjusted. As an experiment, she wants you to raise the prices of all BMW rentals by 2 percent. You explain to Penny that an update query can perform this task.

Because Penny is interested in tracking all information in the database, you suggest two formal reports that could be of use to her. The first is a Customer Rental Summary report that lists each customer and what vehicles they have rented so far. The second report is an Owners' Income report, which will display how much money the owners are bringing in with their rentals. Penny agrees that these reports will be useful for her analysis.

ASSIGNMENT 1: CREATING THE DATABASE DESIGN

In this assignment, you design your database tables using a word-processing program. Pay close attention to the logic and structure of the tables. Do not start developing your Access database in Assignment 2 before getting feedback from your instructor on Assignment 1. Keep in mind that you need to examine the requirements in Assignment 2 to design your fields and tables properly. It is good programming practice to look at the required outputs before beginning your design. When designing the database, observe the following guidelines:

- First, determine the tables you will need by listing the name of each table and the fields it should contain. Avoid data redundancy. Do not create a field if it can be created by a calculated field in a query.
- You will need a transaction table. Think about the business events that occur with each car rental. Avoid duplicating data.
- Document your tables using the table feature of your word processor. Your tables should resemble the format shown in Figure 4-1.
- You must mark the appropriate key field(s) by entering an asterisk (*) next to the field name. Keep in mind that some tables might need a compound primary key to uniquely identify a record within a table.
- Print the database design, if required.

Table
Field
FIGURE 4-1
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